AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 1 · STUDENT LESSON

Requirements and service boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Split the request. Check the boundary. Use the approved evidence.

Trace a read-only service with supported answers, missing-record review and excluded actions.

Keep one habit: Timetable contract → complete decision trace → museum picture → probe table → local simulation → independent inventory.

MAIN LESSON · SECTION 1

Build a service that knows what it may do

> Separate supported information, missing evidence and excluded actions.

Learning goals

I can explain the **mechanism** and terminology of requirements and service boundaries.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve traceable evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Neural network experiment defense. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall AI 5 exit **evidence**, tested Python programs, APIs, structured data, version control and basic system diagrams.. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain requirements and service boundaries using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of requirements and service boundaries

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Turn a broad idea into an observable service contract

> Evidence and permission answer different questions.

Start with the two-part student request

Imagine a school timetable desk with a public printed schedule. A student asks, “When is Tuesday’s meeting, and can you move it to Wednesday?” These are two operations: reading a time and changing a schedule. The desk can answer the supported read while declining the edit. A useful answer does not create permission to change the source. We will model this contract locally, with invented records and no live account.

Define the service boundary

A **service boundary** states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

Complete classroom contract

Contract question	This classroom service
Who requests?	A fictional student; identity/authentication is not implemented here
Approved evidence	Supplied public-schedule-classroom-v1: Tuesday → 16:00
Allowed operation	Read a meeting time from that supplied schedule
Excluded operations	Move/edit meetings; send email; any other action

Contract question	This classroom service
Read with no supplied entry	Return review status, without inventing a time
Mixed request	Return a separate outcome for each operation
Observable requirements	Supported read names its source; excluded actions do not execute; schedule stays unchanged

Record supplied and missing facts

> **List the given facts and one fact that is not supplied.**

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace permission first, then look up allowed information

> A request can contain both an answerable read and an excluded action.

Original timetable service definition

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Complete three-outcome trace

Split the request before deciding each operation

OPERATION	PERMISSION AND EVIDENCE CHECK	OUTCOME
Read Tuesday time	Read allowed; Tuesday entry says 16:00	answered: Tuesday 16:00, source public-schedule-classroom-v1
Move Tuesday meeting to Wednesday	Edit is outside the contract	out_of_scope: no change executed
Read Wednesday time, a separate probe	Read allowed; Wednesday entry absent	review: no supplied entry, no guessed time

One allowed operation does not authorize another. Missing evidence and excluded operations receive different outcomes.

Explain the checks and evidence limits

Follow the checks in order. First name the operation. If it is excluded, stop that operation without changing the schedule. If a read is allowed, look for the required approved record. Return the supplied time with its source when present; otherwise return review. In this local exercise review is a status to record, not proof that a person has checked or resolved the case.

The time has no supplied date or time zone, so do not present it as a verified real appointment.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture to the deciding step

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Requirements and service boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.	The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.	Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the service boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one part of the request and predict each outcome

> Keep the read-only boundary fixed.

Current task and response

- A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the missing-card picture

- Change: Replace P with Q, which has no approved card.

▼ **Check after predicting**

Prediction: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare the timetable with a museum fact desk

> Use the same distinction between evidence and permission in a different setting.

Complete main request and answer

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.

Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Explain the scope of this trace

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate museum case and all givens

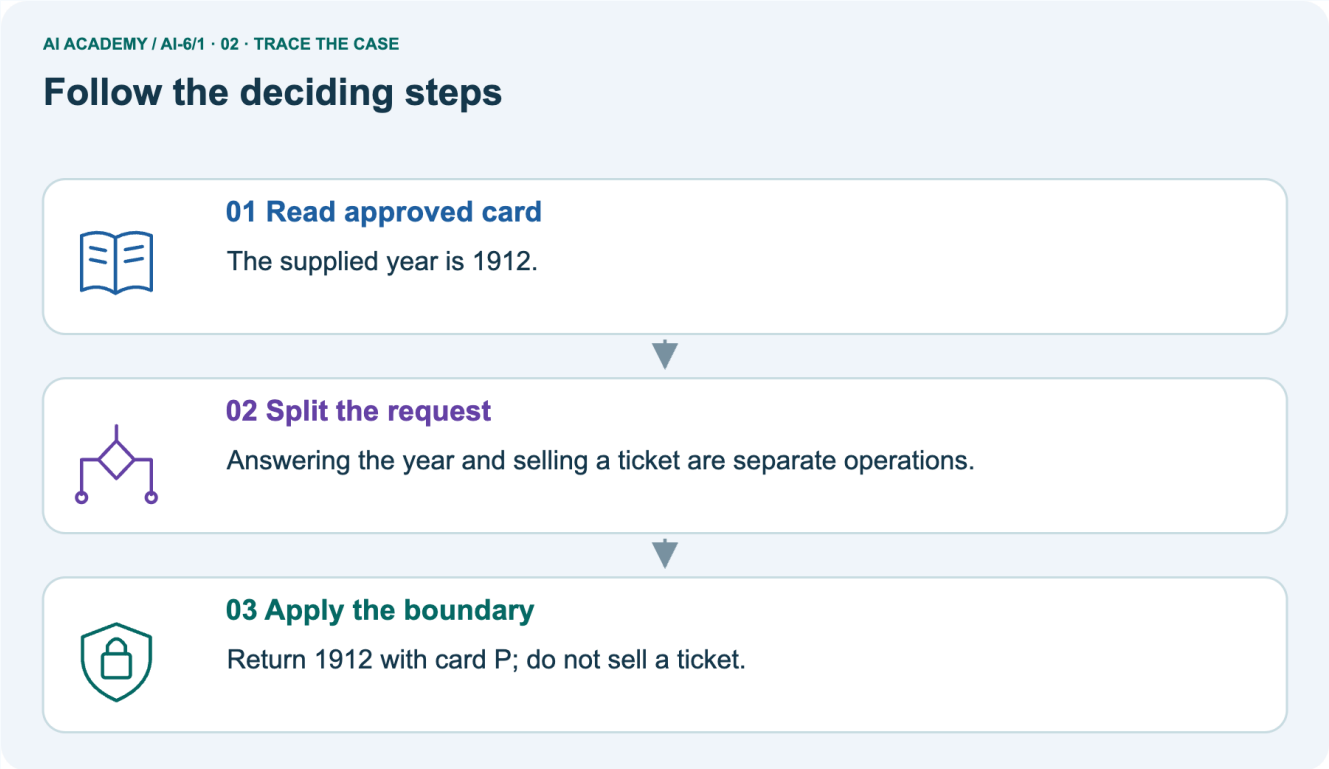
Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

Approved card P says 1912. The contract permits reading cards, not selling tickets.

The main case is a timetable; the picture is a separate museum fact desk. Approved card P supplies year 1912. The contract allows reading cards but excludes selling tickets. Return the supplied year with card P and decline the ticket sale. For Q with no approved card, the year is unsupported and needs review; selling a ticket remains excluded.

No real museum history, ticket system or human review outcome has been checked in this example.

Trace card lookup and excluded sale



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read approved card	The supplied year is 1912.
2. Split the request	Answering the year and selling a ticket are separate operations.
3. Apply the boundary	Return 1912 with card P; do not sell a ticket.

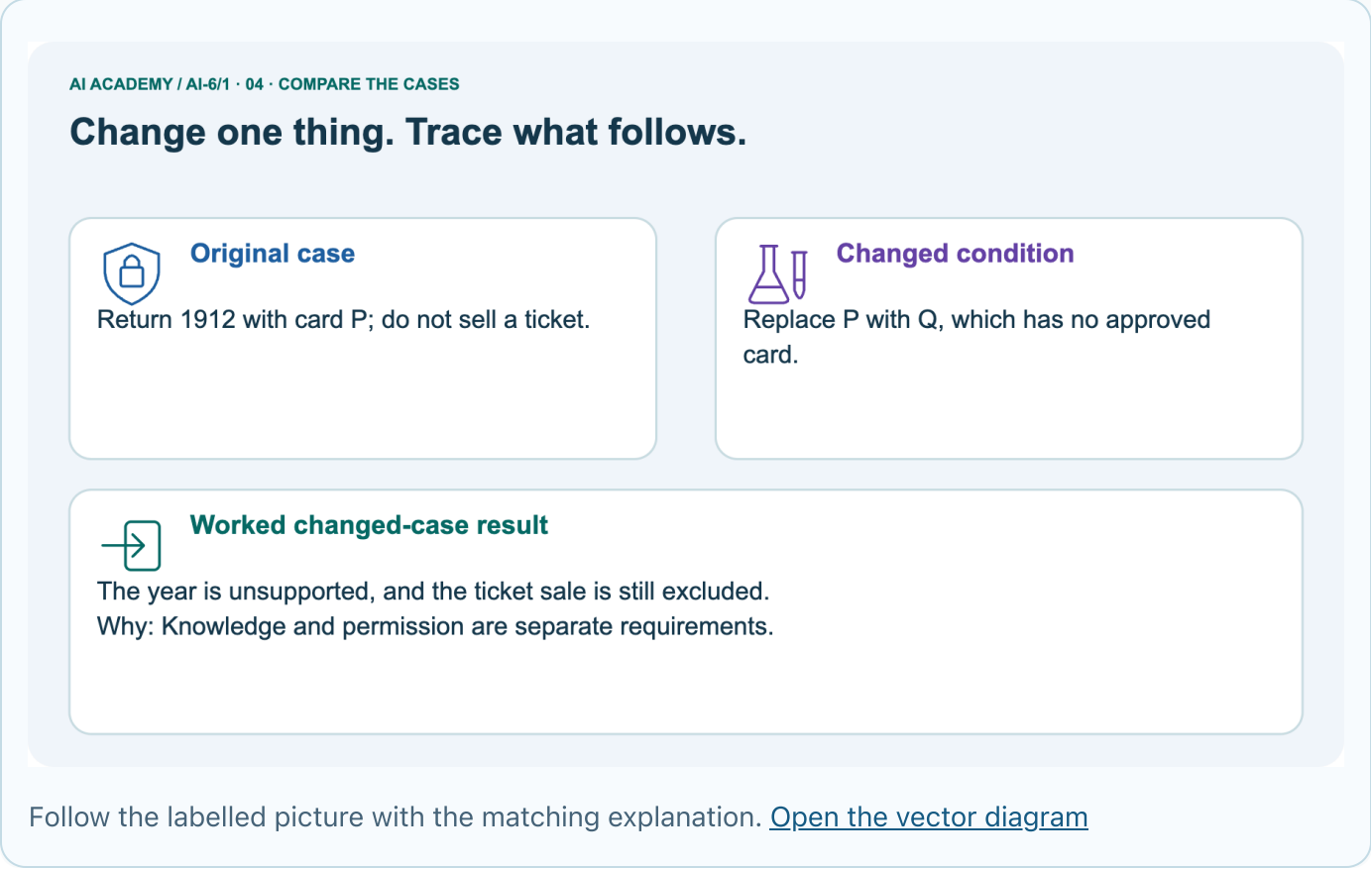
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **service boundary** states permitted work; useful information does not grant extra operations.

Predict replacing P with missing-card Q

- Change one thing: Replace P with Q, which has no approved card. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain both operations and the history limit

- An approved card can be outdated; this trace does not independently verify museum history.
- Read the labelled picture: Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

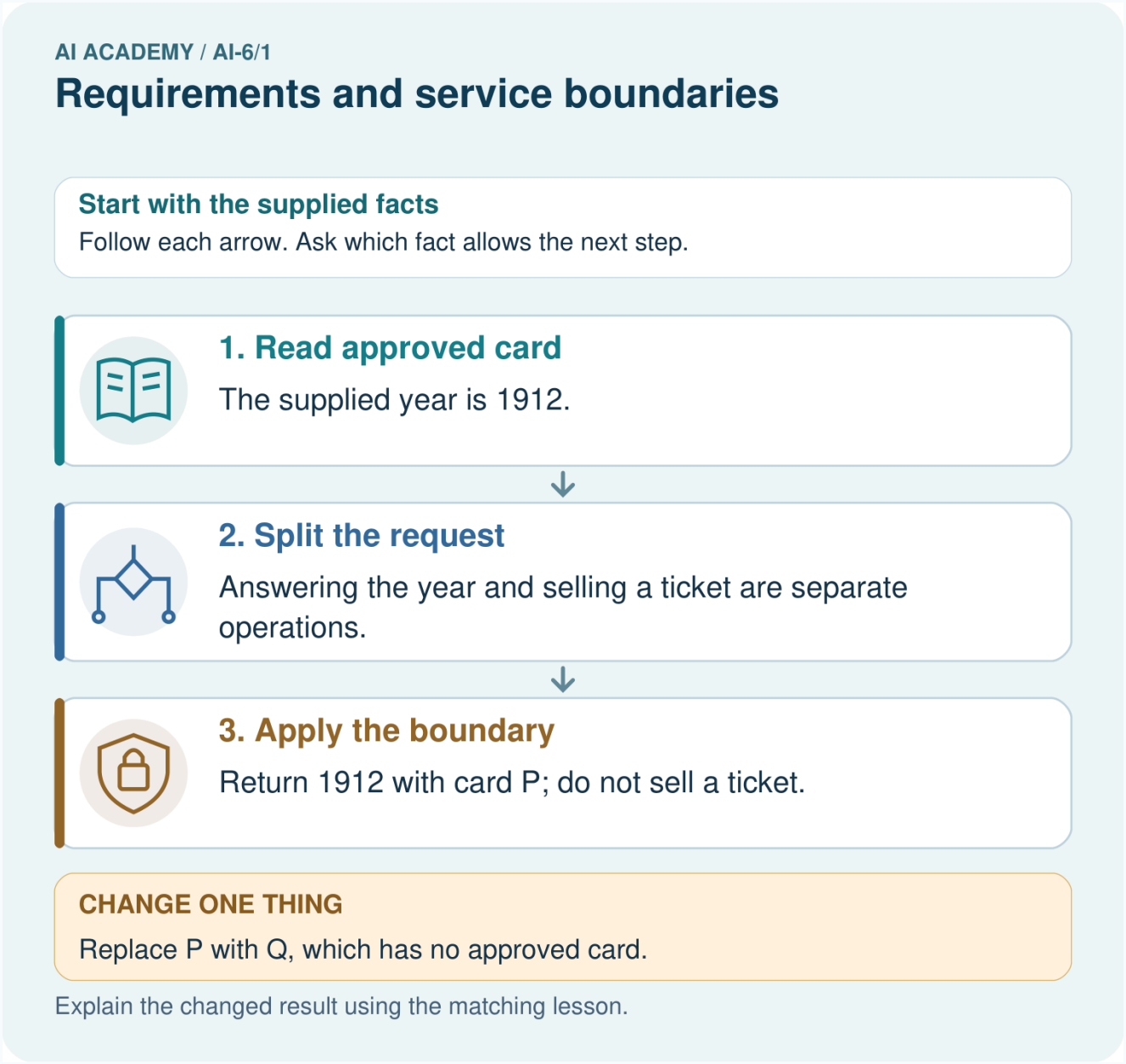
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Requirements and service boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Diagnose a wrong answer, a missing record and an excluded action

> These conditions need different responses.

Core boundary qualification and error analysis

A helpful answer does not grant an operation that the service contract excludes.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete normal, missing and boundary probes

Complete probe	Expected outcome and reason
Read Tuesday; then move it	Answer 16:00 from source; reject move; schedule unchanged
Read Wednesday, no supplied entry	review; do not invent a time
Send email	out_of_scope; no email function exists in the simulation
Unknown structured operation	out_of_scope under the explicit read-only allowlist
Read action with missing day field	Invalid structured request; complete the input before a lookup
A newer conflicting schedule appears	Current one-source exercise supplies no conflict-resolution rule; review/version the contract before extending it

Complete probe	Expected outcome and reason
Return review status	Records an unresolved case; no reviewed decision is implied

Guided explanation and repair

- A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise an operation-by-operation explanation

> Name the request, approved evidence, decision and outcome.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Requirements and service boundaries”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Independent reasoning check

- Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Museum picture response

- Split the museum visitor’s request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

> **My explanation using a pictured step and the given facts**
>
> _____
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Reserve the independent inventory case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run a complete local boundary simulation

> Check the outcome and the unchanged source, not only a polite refusal sentence.

Current worked-request investigation

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

Complete runnable dispatcher and invariant checks

```
"""Local deterministic simulation of a read-only timetable service."""
from copy import deepcopy
SCHEDULE={'Tuesday':'16:00'}
SOURCE='public-schedule-classroom-v1'

def dispatch(operations,schedule):
    if not isinstance(operations,list) or not operations:
        raise ValueError('Need a nonempty list of structured operations')
    results=[]
    for operation in operations:
        if not isinstance(operation,dict) or not isinstance(operation.get('action'),str):
            raise ValueError('Every operation needs a string action')
        action=operation['action']
        # The contract only authorizes reads; requests cannot extend this allowlist.
        if action!='read_schedule':
            results.append({'action':action,'status':'out_of_scope'})
            continue
        day=operation.get('day')
        if not isinstance(day,str) or not day:
            raise ValueError('A schedule read needs a nonempty day string')
        if day not in schedule:
            results.append({'action':action,'day':day,'status':'review',
                            'reason':'No supplied schedule entry'})
        else:
            results.append({'action':action,'day':day,'status':'answered',
                            'time':schedule[day],'source':SOURCE})
    return results

before=deepcopy(SCHEDULE)
request=[{'action':'read_schedule','day':'Tuesday'},
         {'action':'move_schedule','day':'Tuesday','new_day':'Wednesday'}]
result=dispatch(request,SCHEDULE)
```

```
assert result[0]['time']=='16:00' and result[1]['status']=='out_of_scope'
assert dispatch([{'action':'read_schedule','day':'Wednesday'}],SCHEDULE)[0]
['status']=='review'
assert dispatch([{'action':'email_user'}],SCHEDULE)[0]['status']=='out_of_scope'
assert SCHEDULE==before
print('Mixed request:',result)
print('Missing
Wednesday:',dispatch([{'action':'read_schedule','day':'Wednesday'}],SCHEDULE))
print('Snapshot unchanged:',SCHEDULE==before)
print('Simulation only: no natural-language parser, network, authentication, email, edit
or reviewer outcome.')
```

Expected cases and simulation limits

1. Predict a separate status for each operation before running or tracing. The simulation consumes structured action dictionaries, not natural-language requests. Check answered Tuesday, excluded move/email, and review for missing Wednesday. Compare the schedule with its saved snapshot to confirm no edit occurred. Missing required fields are invalid input, distinct from a valid lookup with no record.

   The example has no network, authentication, natural-language parser, actual editing, email delivery or completed human review. It demonstrates a deterministic contract before adding model behaviour; it does not prove a deployed service is secure or correct.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: apply the boundary to a read-only inventory

> Use the new source and operations before opening feedback.

Complete independent inventory case

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save a small service contract and its acceptance cases

> Keep inputs, approved source, excluded actions and observed results together.

Project task and artifact

Add a labelled requirements and service boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of requirements and service boundaries.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a supported answer and an enforced boundary

> A correct response should say what was done and what remains unresolved.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of requirements and service boundaries

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Service boundary	The declared inputs, data and operations of a service
Acceptance criterion	An observable requirement
Out of scope	Not authorised by this service contract

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Requirements and service boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.	The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.	Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the service boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

By the end, I can	How I will show it
Explain the mechanism and terminology of requirements and service boundaries	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.
- Which labelled step matches this detail? Return 1912 with card P; do not sell a ticket.
- 1. Read approved card
- 2. Split the request
- 3. Apply the boundary
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply the boundary Return 1912 with card P; do not sell a ticket. A **service boundary** states permitted work; useful information does not grant extra operations.

Changed-case question and optional hint

- Changed-case practice: Replace P with Q, which has no approved card. Which conclusion is supported by the supplied facts?
- 1. The year is unsupported, and the ticket sale is still excluded.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The year is unsupported, and the ticket sale is still excluded. Knowledge and permission are separate requirements.

Repair and retain the record

Review Neural network experiment defense (ai-5/36). Goal: Defend a network experiment using traceable **evidence**. Short practice: Calculate both weighted terms, z and ReLU **output**.

Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

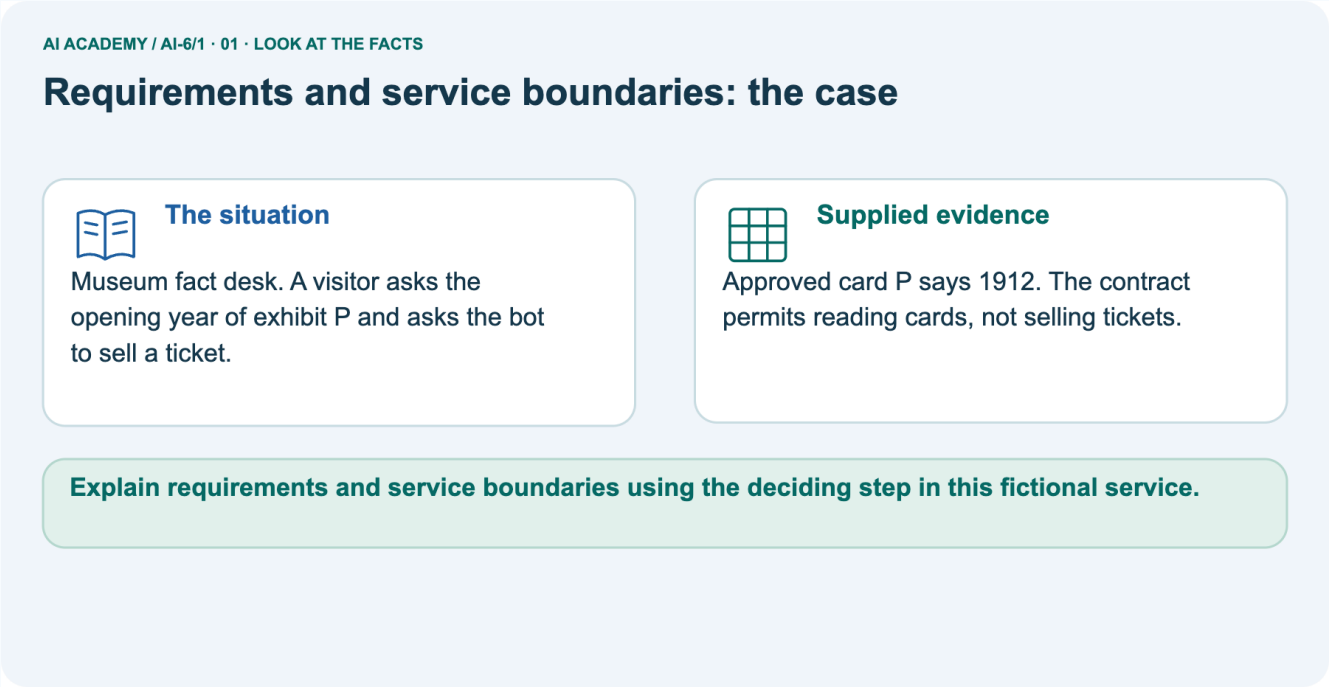
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Requirements and service boundaries: the case

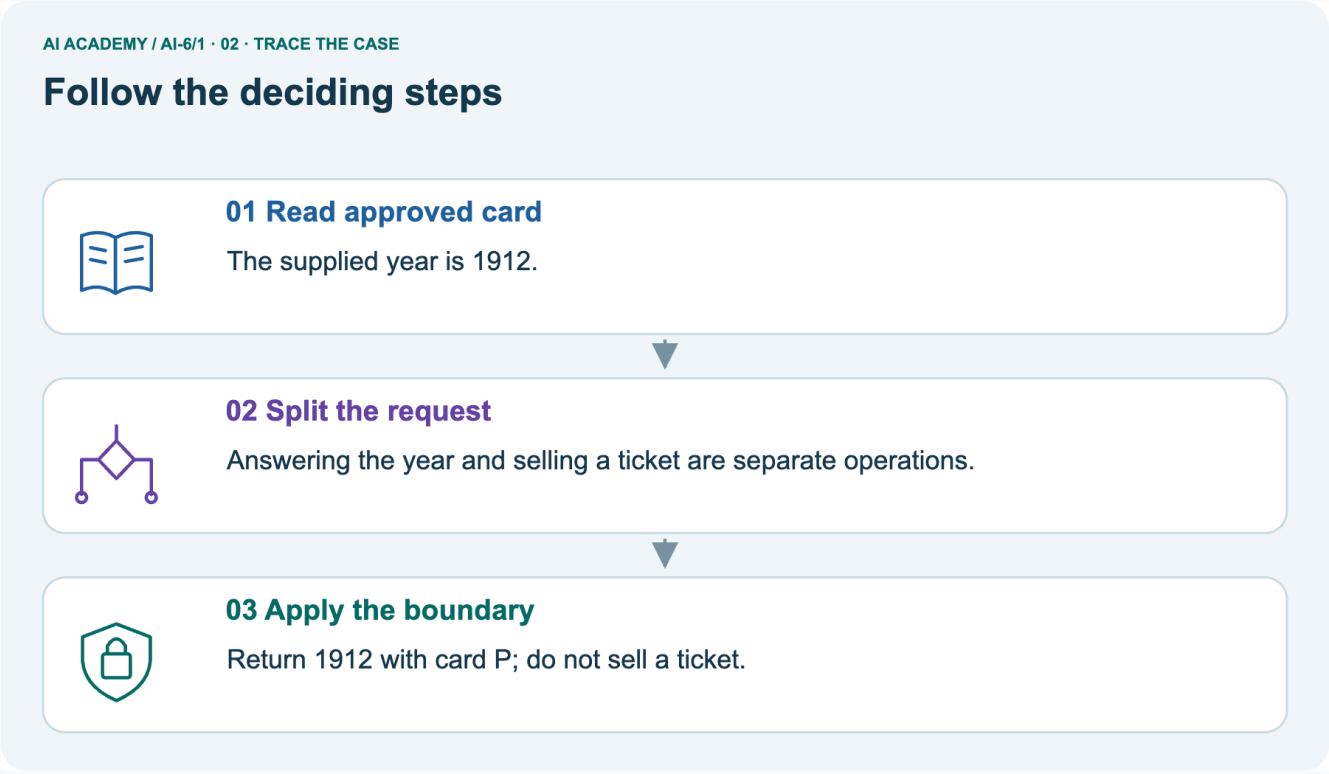


Follow the labelled picture with the matching explanation. Open the vector diagram

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket. Supplied facts: Approved card P says 1912. The contract permits reading cards, not selling tickets.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

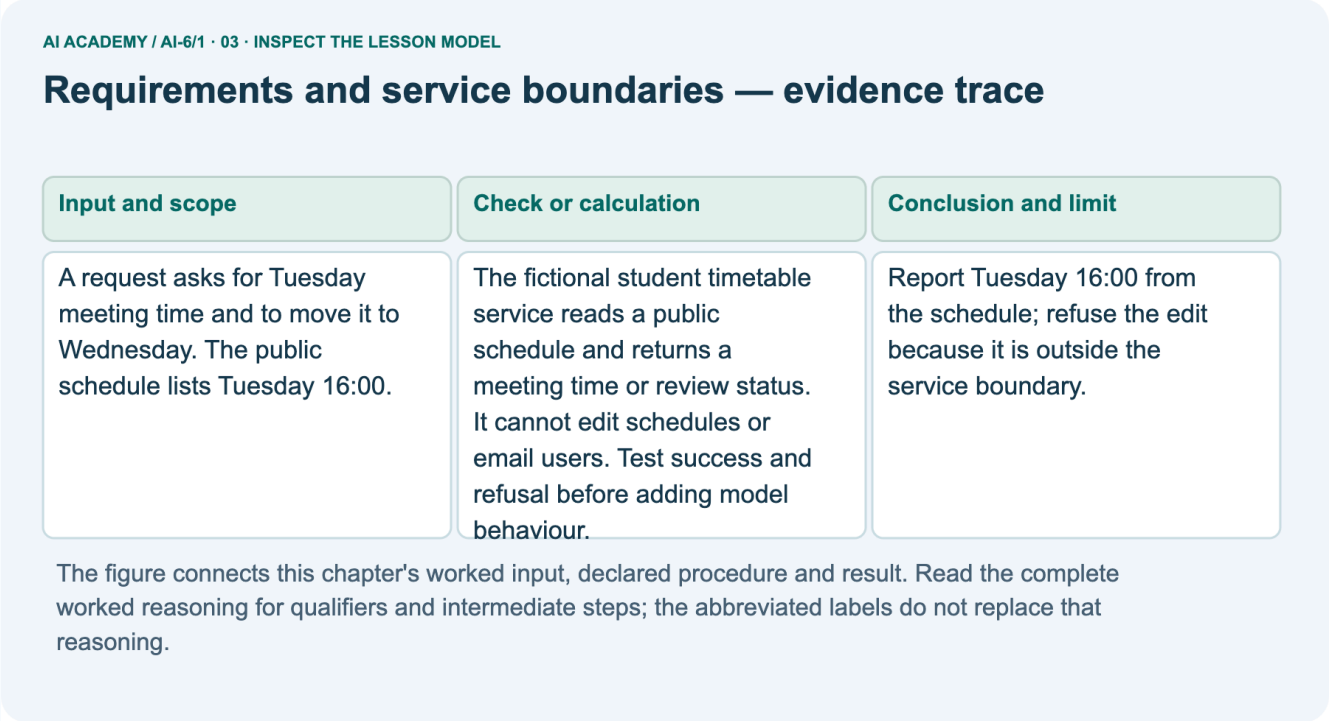


Follow the labelled picture with the matching explanation. Open the vector diagram

Read approved card: The supplied year is 1912. → Split the request: Answering the year and selling a ticket are separate operations. → Apply the boundary: Return 1912 with card P; do not sell a ticket.

Point to each numbered stage and explain which recorded **input** it uses.

03. Requirements and service boundaries — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

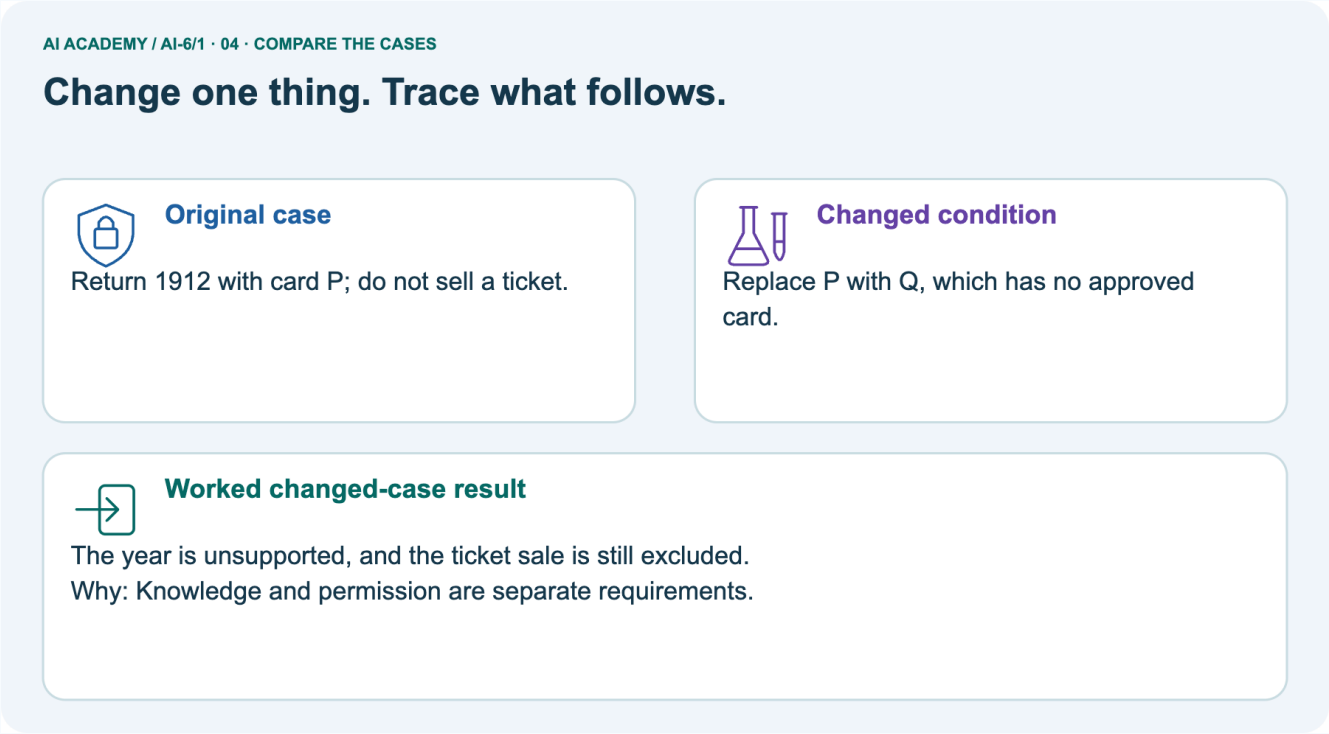
Requirements and service boundaries — **evidence** trace. A request asks for Tuesday meeting time and to move it to Wednesday.

The public schedule lists Tuesday 16:00. | The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users.

Test success and refusal before adding model behaviour. | Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



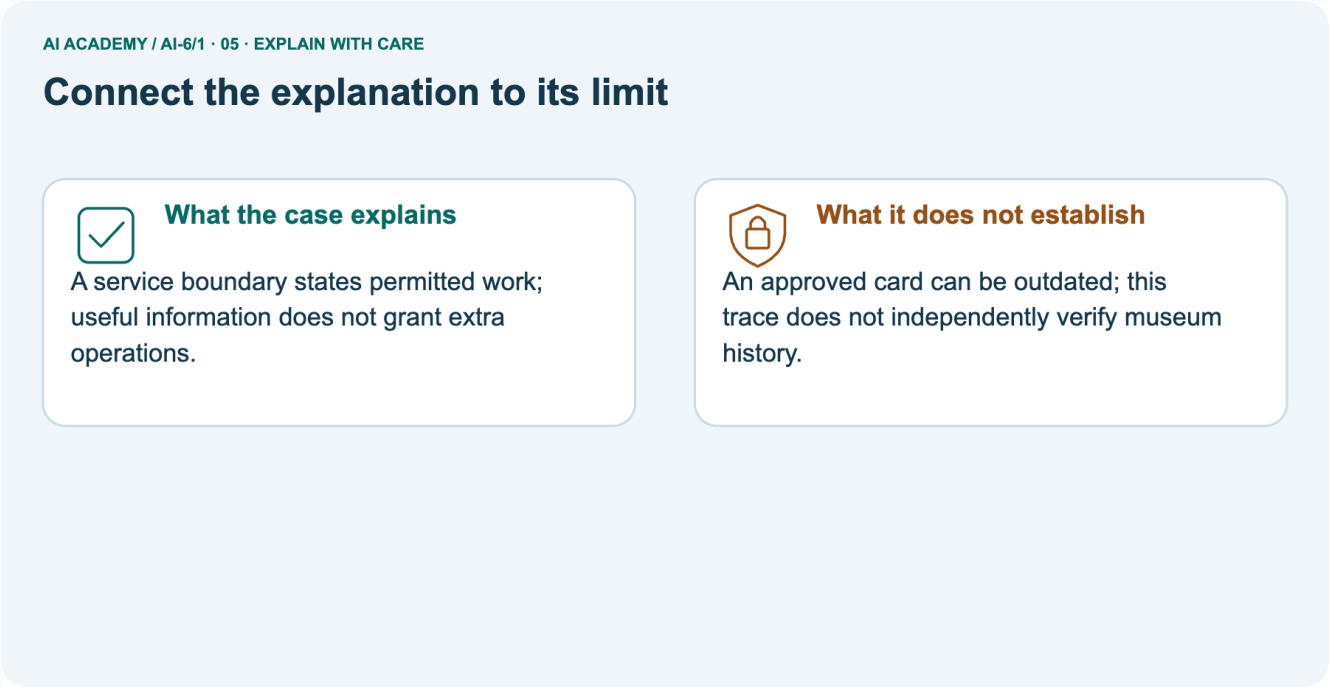
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return 1912 with card P; do not sell a ticket. Changed condition: Replace P with Q, which has no approved card. Expected result: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

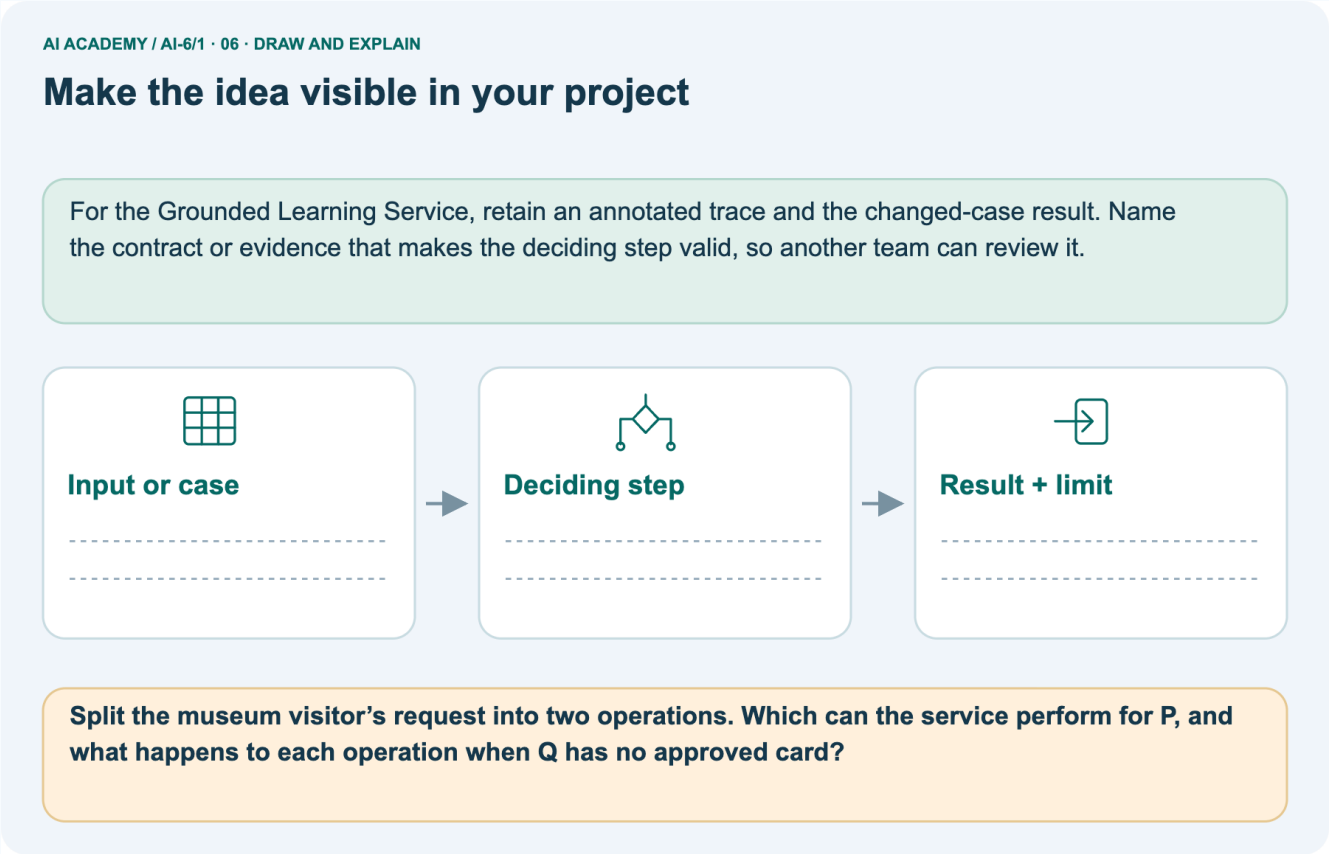


Follow the labelled picture with the matching explanation. Open the vector diagram

A **service boundary** states permitted work; useful information does not grant extra operations. Limit: An approved card can be outdated; this trace does not independently verify museum history.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Split the museum visitor’s request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Requirements and service boundaries: the case

02. Follow the deciding steps

03. Requirements and service boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.